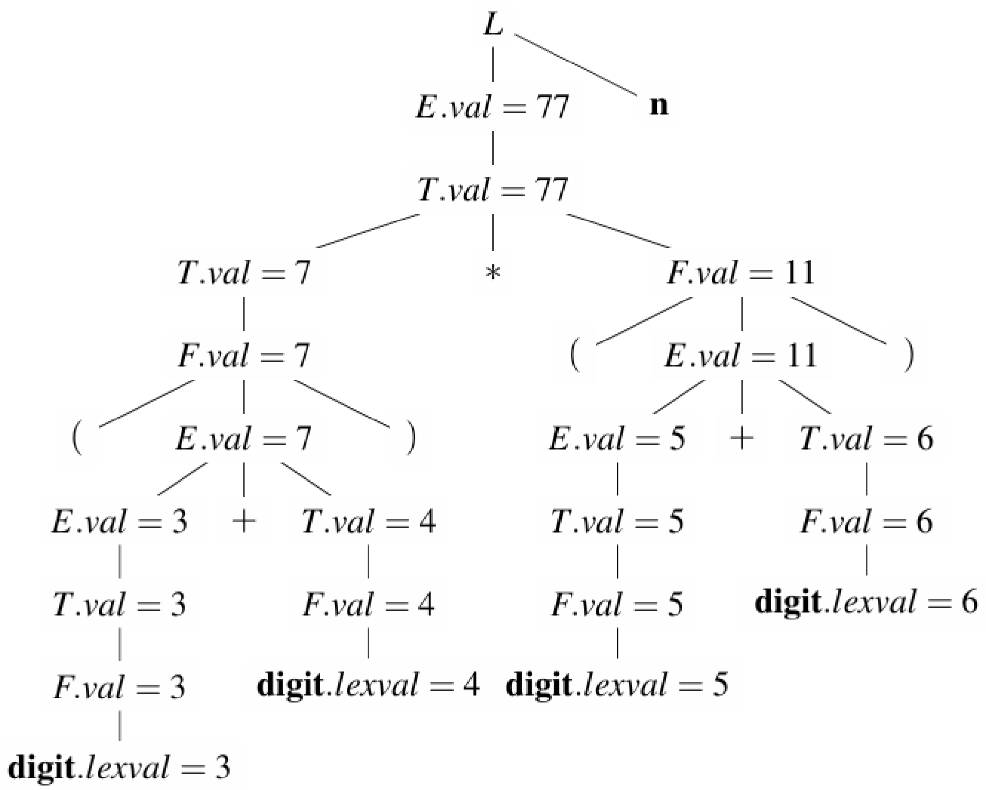
Assignment 5

Name:

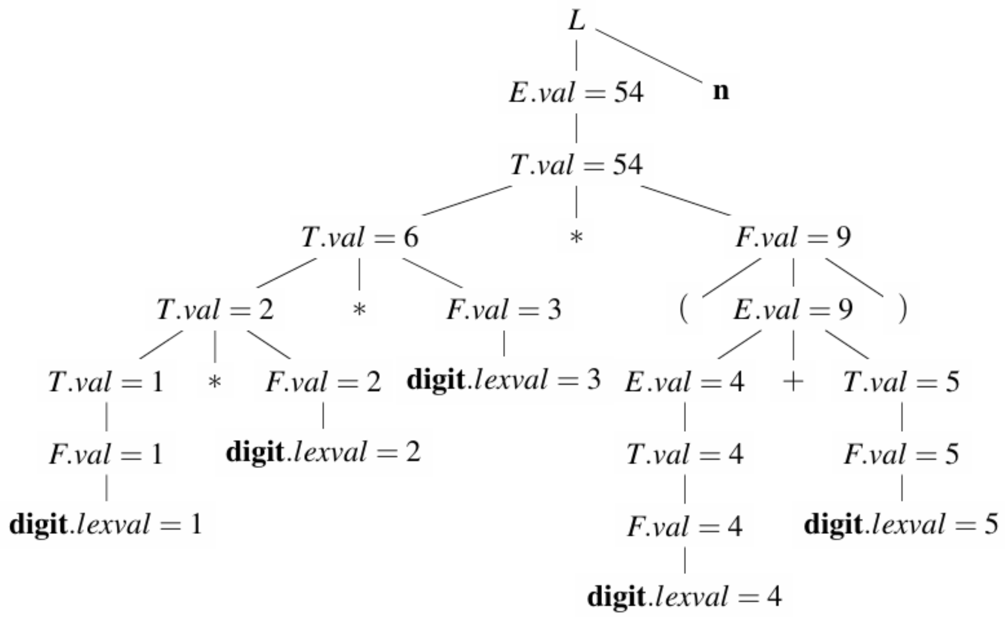
1 Required Exercises

Exercise 1

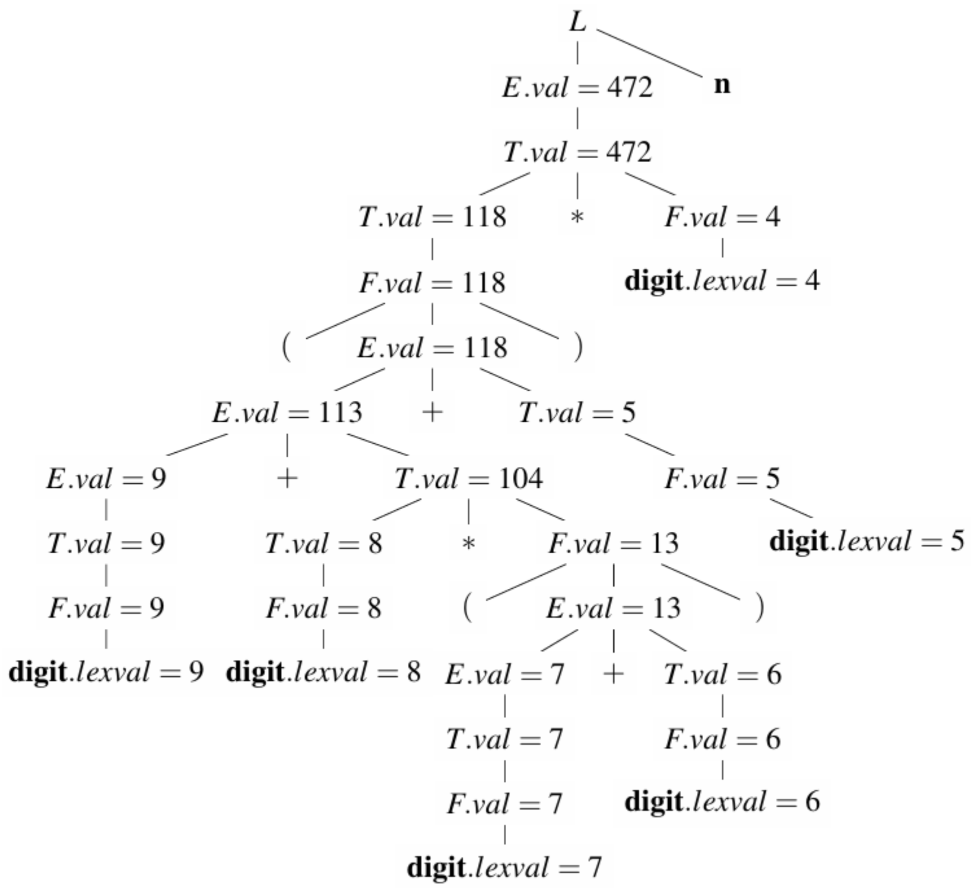
1. $(3+4)*(5+6)n$



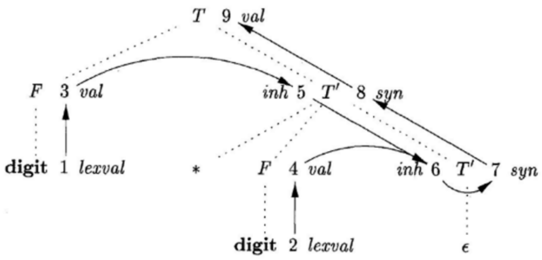
2. $1 * 2 * 3 * (4 + 5)n$



3. $(9 + 8 * (7 + 6) + 5) * 4n$



Exercise 2



The productions and semantic rules for the grammar involved in figure are as follows

	PRODUCTION	SEMANTIC RULES
1)	$T \to F\,T'$	$T'.inh = F.val$ $T.val = T'.syn$
2)	$T' \to *\,F\,T'_1$	$T'_1.inh = T'.inh \times F.val$ $T'.syn = T'_1.syn$
3)	$T' \to \epsilon$	$T'.syn = T'.inh$
4)	$F \to$ **digit**	$F.val =$ **digit**.$lexval$

Based on the analysis, we can derive the sequence of attribute computation as follows, where (1, 3) indicates that step 1 must precede step 3.

(1, 3) (2, 4) (3, 5) (5, 6) (4, 6) (6, 7) (7, 8) (8, 9)

There all totally **10** different topological sorts for the dependency graph.

1	(1, 2, 3, 4, 5, 6, 7, 8, 9)
2	(1, 2, 3, 5, 4, 6, 7, 8, 9)
3	(1, 2, 4, 3, 5, 6, 7, 8, 9)
4	(1, 3, 2, 4, 5, 6, 7, 8, 9)
5	(1, 3, 2, 5, 4, 6, 7, 8, 9)
6	(1, 3, 5, 2, 4, 6, 7, 8, 9)
7	(2, 1, 3, 4, 5, 6, 7, 8, 9)
8	(2, 1, 3, 5, 4, 6, 7, 8, 9)
9	(2, 1, 4, 3, 5, 6, 7, 8, 9)
10	(2, 4, 1, 3, 5, 6, 7, 8, 9)

Exercise 3

PRODUCTIONS	SEMANTIC RULES
1) $E \to E1 + T$	$E.type = float\ if\ E1.type = float\ or\ T.type = float\ , else\ int$
2) $E \to T$	$E.type = T.type$
3) $T \to num.num$	$T.type = float$
4) $T \to num$	$T.type = int$